Mid-term Exam Review

Science & Technology Innovation Management

Lecturer: Dr. Do Thi Hai | USTH, Master Programme

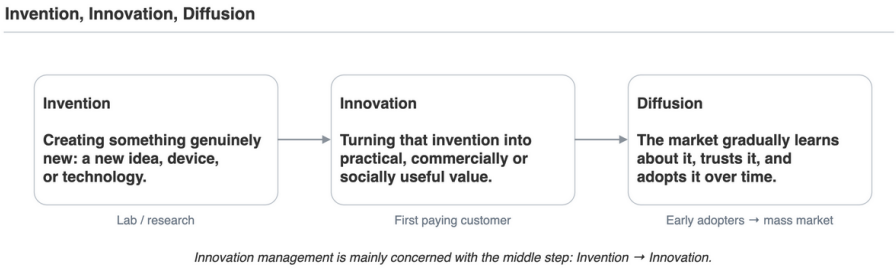
Exam: 23/10/2026, 14:55–15:55 (written, 1 hour)

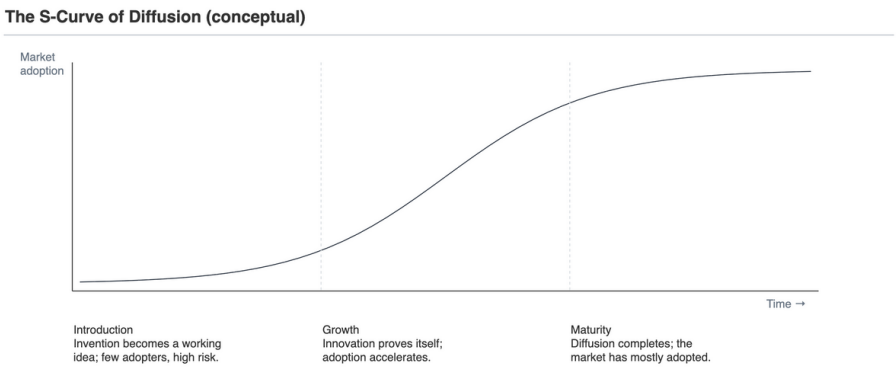
1. Definition of innovation

"Definition of innovation."

OECD (2015): "the implementation of a new or significantly improved product, process, marketing method, or organizational method." **Tidd & Bessant (2020):** "turning opportunity into new ideas and putting these into widely used practice" (Innovation = Conception + Invention + Commercial Exploitation). Both agree: an idea only counts once *implemented and used*, not merely conceived.

- **Invention** – proof of concept: a new device, method, or process based on scientific knowledge, usually stopping at the lab or blueprint stage.
- **Innovation** – commercializing that invention: turning it into something the market actually buys or a real problem it actually solves. An invention never brought to market is not yet an innovation under either definition above.
- **Diffusion** – the market gradually learns about it, trusts it, and adopts it, spreading from early users to the mass market.
- Innovation management focuses mainly on the **middle step**: organizing people, resources, and decisions so ideas survive the journey.





2. The "4Ps" model (Tidd & Bessant)

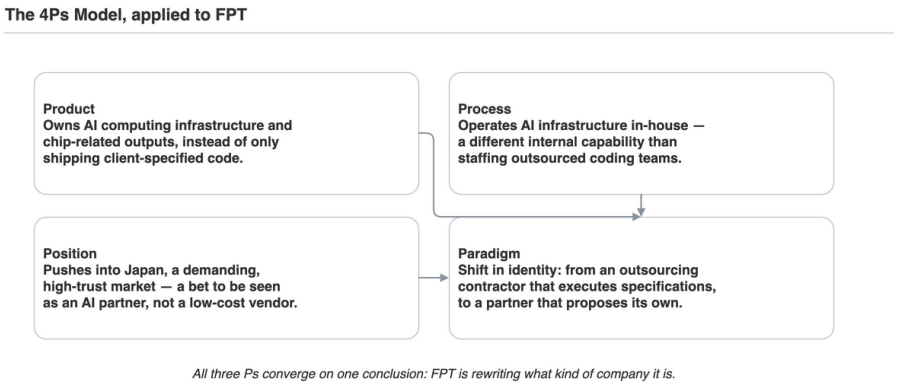
"What does the "4Ps of Innovation" model by Tidd & Bessant entail? Provide an example of how the innovation strategies of Product, Process, Position, and Paradigm are applied at a major technology company (such as Apple, Viettel, or FPT)."

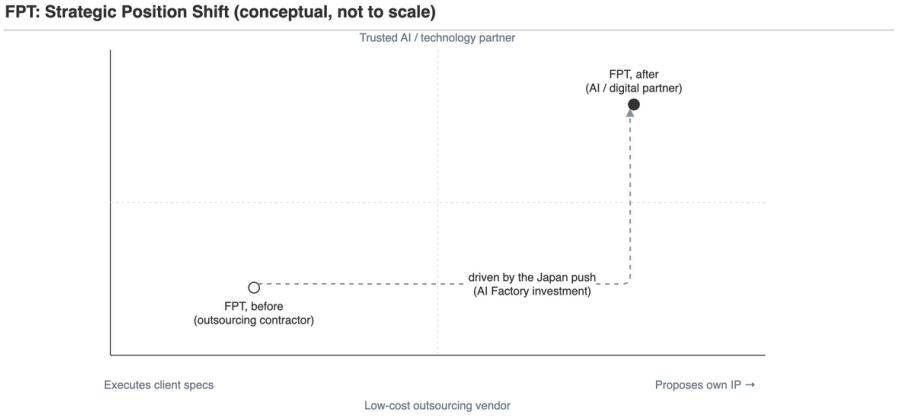
Tidd & Bessant (2020) identify four dimensions along which a company can innovate, often at the same time – not just by inventing a new gadget:

- **Product** – what the organization offers.
- **Process** – how it is made and delivered.
- **Position** – who it is sold to, how it competes.
- **Paradigm** – the deepest and rarest: a change in the underlying business model, in what the organization fundamentally is.

FPT is the sharpest case: all four Ps point in one direction, away from being hands for hire, toward owning its own IP.

- **Product** – builds and owns AI computing infrastructure and chip-related offerings, instead of only shipping client-specified software.
- **Process** – operating AI infrastructure in-house needs a different internal capability than staffing outsourced coding teams.
- **Position** – pushes into Japan, a demanding, high-trust market, betting on being seen as an AI partner, not a low-cost vendor.
- **Paradigm** – the conclusion the other three drive toward: from a contractor executing someone else's specification, to a partner proposing its own.





3. Risks in the "Implementing" phase

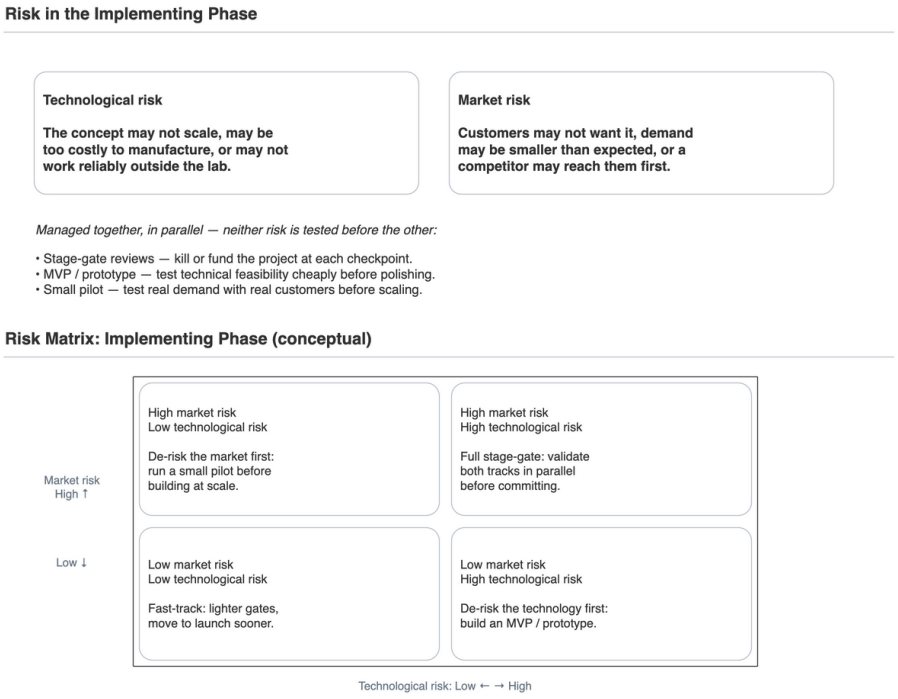
"During the "Implementing" phase, what risks typically arise when transforming a technical concept into a commercial product or service? How can one manage both technological and market risks?"

In **Tidd & Bessant's Search–Select–Implement–Capture model**, "Implement" is where a screened idea is finally committed to real resources (R&D, prototyping, commercial launch) – uncertainty is still high, but the cost of being wrong rises sharply.

- **Technological risk** – the idea may not scale, may cost far more to manufacture than expected, or may behave unpredictably outside the lab.
- **Market risk** – a technically sound product may still fail commercially: customers do not want it as assumed, demand is smaller than projected, or a rival reaches customers first.

Managing both means learning in stages, not committing all resources at once:

- **Stage-gate** reviews – stop a weak idea early, rather than discover it weak only after full launch.
- **Minimum viable product** – test the core technical idea cheaply before polishing it.
- **Small pilot** – test real demand with real customers.
- Critically, run both tracks **in parallel, not sequentially**.



4. The role of innovation management (5 official roles)

"Analyze the role of innovation management for businesses. Provide illustrative examples."

The course defines innovation management through five official roles – each shown here with how **Grab** fills it:

- **(1) Competitive advantage** – an edge made repeatable, not a one-time success. *Grab:* continuous, data-driven experimentation in matching and pricing.
- **(2) Integration** – technology, market, and organization pulling the same direction. *Grab:* the super-app, bundling rides, delivery, and payments into one platform.
- **(3) Ideas into real value** – the harder work of shipping past the idea stage. *Grab:* its

map/location-data tool, built internally, later sold externally.

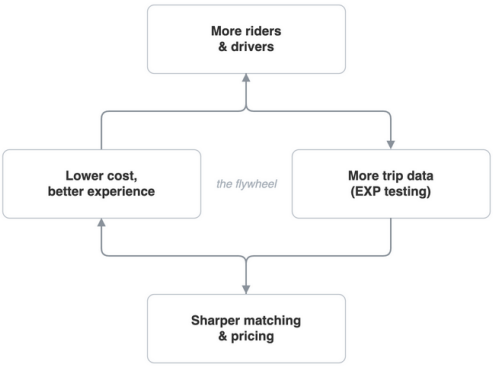
- **(4) Dynamic capabilities** – the muscle to adapt as conditions shift. *Grab:* a disciplined, gradual cut in promotional spending as it matured.
- **(5) Risk & resources** – expanding responsibly without exposing the core business. *Grab:* a dedicated fraud-defense system, and licenses secured before lending.

Five Roles of Innovation Management, Grab

1. Competitive advantage	2. Integration	3. Ideas → value	4. Dynamic capabilities	5. Risk & resources
Continuous A/B testing keeps matching and pricing sharp.	Super-app: rides, delivery and payments fused into one loop.	Internal map tool later sold as a B2B product.	Disciplined cut in promo spend as the business matured.	Fraud-defense system; licenses secured before launching lending.

Each role is a lens on the same company, not five separate companies.

Grab's Reinforcing Flywheel (conceptual, Roles 1 & 2)



Each turn of the loop makes the next turn cheaper and faster to win — sustaining competitive advantage.

Good luck this afternoon, 14:55–15:55!